

Seat No: _____

Enrollment No: _____

PARUL UNIVERSITY
FACULTY OF IT & COMPUTER SCIENCE
MCA/M.Sc.IT 2025-26 Mid-Term Examination

Semester: 03

Date: 06-08-2025

Subject Code: 05201350

Time: (1hr:30min)

Subject Name: Artificial Intelligence - II

Total Marks: 40

Instructions:

1. Figures to the right indicate full marks.
2. Make suitable assumptions wherever necessary.

Q.1	Answer the following.	CO	PO	BL	[10]
(a)	3 short questions of 1 mark each				[3]
	(i) Define <i>Rote Learning</i> in Artificial Intelligence.	01	01	Remembering (BT-1)	
	(ii) What is the goal of a planning system in AI?	02	01	Remembering (BT-1)	
	(iii) What is the range of values for a membership function in fuzzy logic?	02	01	Remembering (BT-1)	
(b)	Objective type/MCQs/True-False/Fill in blanks (7 questions of 1 mark each)				[7]
	1 Which type of learning is MOST evident in Google Duplex when it refines its responses by using large datasets of recorded conversations? (A) Rote Learning (B) Inductive Learning (C) Learning from Examples (D) Learning by Taking Advice	01	01	Remembering (BT-1)	
	2. Alpha-beta pruning always changes the final minimax value of the game tree: True/False Justify your answer	02	01	Understanding (BT-2)	
	3. Which search algorithm is commonly used in game playing to decide optimal moves in two-player games?	02	01	Remembering (BT-1)	
	4. The process of converting fuzzy values back into a crisp, actionable value is called _____.	02	01	Remembering (BT-1)	
	5. How does <i>learning by taking advice</i> benefit AI systems like Google Duplex?	01	01	Understanding (BT-2)	
	6. What is fuzzification in a fuzzy control system?	02	01	Remembering (BT-1)	
	7. How do linguistic variables enhance reasoning in fuzzy logic systems?	02	01	Understanding (BT-2)	
Q.2	Answer the following. (2 or 3 mark questions)				[10]
(a)	Two Questions of 2 Marks				[4]
	(i) Explain how inductive learning can be used to enhance the conversational abilities of Google Duplex.	01	01	Understanding (BT-2)	(02)
	(ii) Why are membership functions important in fuzzy logic systems?	02	01	Understanding (BT-2)	(02)
(b)	Two Questions of 3 Marks				[6]
	(i) Explain the difference between fuzzification and defuzzification with a simple example.	02	01	Understand/Apply (BT-2/3)	(03)
	(ii) Consider a tic-tac-toe game tree where Alpha-Beta pruning is applied. Explain how alpha-beta pruning helps improve the efficiency of the minimax search and what remains unchanged after pruning.	02	01	Apply/Analyse (BT-3/4)	(03)
Q.3	Attempt any TWO.				[10]
	(i) Using the Blocks World example, describe how a Goal Stack Planning system would plan to move block A from the top of block B to the table. Include the steps the planner would generate.	02	01	Apply (BT-3)	(05)
	(ii) Analyze how different AI learning types (Rote Learning, Learning from Examples, Learning by Taking	02	01	Analyse (BT-4)	(05)

	Advice, Learning in Problem Solving) contribute to the development of Google Duplex. Provide relevant examples for each learning type.				
	(iii) Evaluate the use of fuzzy rule-based systems in real-world control applications (e.g., washing machines, climate control). Discuss their advantages and any limitations.	02	01	Evaluate (BT-5)	(05)
Q.4	Answer the following.				[10]
(a)	A smart air conditioning system uses fuzzy logic to adjust room temperature. The system takes two inputs – “Room Temperature” (Cold, Comfortable, Hot) and “Humidity” (Low, Medium, High). Based on these, it controls the “Fan Speed” (Low, Medium, High). Analyze how the system would respond when the room temperature is “Hot” and humidity is “High.” List the possible fuzzy rules, describe how fuzzification and defuzzification would occur, and explain the final output action.	02	01	Apply/Analyze (BT-3/4)	(05)
(b)	Compare and analyze the effectiveness of Minimax and Alpha-Beta pruning for decision-making in competitive games. In your answer, discuss the trade-off between computation time and search depth.	02	01	Analyze	(05)
	OR				
(b)	Design a simple goal stack plan for a robot to set a table for dinner (tasks: place plate, cup, and spoon). List the main goals, subgoals, and actions the planner would generate.	02	01	Create (BT-6)	(05)